Waves III: Specific Waves

Sound waves are covered in chapter 19 of Halliday and Resnick, while light waves are covered in chapters 39 and 44. For more about light waves, see chapter 9 of Purcell. For water waves, and many other neat wave phenomena, see chapter I-51 of the Feynman lectures and chapter 7 of Crawford's *Waves*. For more about polarization, see chapter I-33 of the Feynman lectures, or for more detail, chapter 8 of Hecht's *Optics*. Basic geometric optics is covered in chapter 40 of Halliday and Resnick, and chapter 1 of Wang and Ricardo, volume 2. There is a total of **82** points.

1 Sound and Longitudinal Waves

Idea 1: Doppler Effect

Working in one dimension with speed of sound c, if a source of sound at frequency f_0 travels at velocity v_s while an observer to their right travels at velocity v_o, the observed frequency is

$$f = \frac{c - v_o}{c - v_s} f_0.$$

Example 1

A speaker is between two perfectly reflective walls and emits a sound of frequency f_0. If you carry the speaker and walk with small speed v towards one of the walls, what do you hear?

Solution

Walking is slow, so we work to lowest order in v/c. The wall you're walking toward experiences a sound of frequency $f_0(1 + v/c)$ by the Doppler effect, and this is the frequency it reflects. Since you're walking towards the wall, a second Doppler effect occurs, causing you to hear frequency $f_0(1+2v/c)$. (We also saw this "double Doppler shift" back in **R1**.)

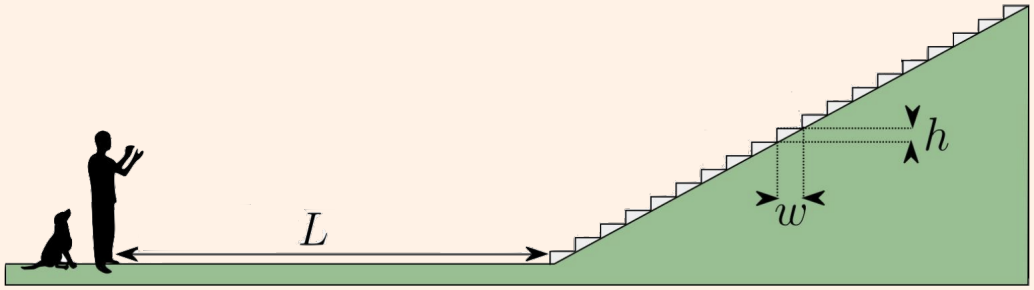
By similar reasoning, you hear sound of frequency $f_0(1 - 2v/c)$ from the wall behind you. And of course, you also hear f_0 from the speaker itself. Thus, in total you hear beats of frequency $2vf_0/c$, as well as a weaker beat of frequency $4vf_0/c$ from the two reflected waves.

Example 2

In my former college at Oxford, there is a long staircase that is said to "quack" when one claps at it. What is the explanation of this phenomenon?

Solution

A diagram of the staircase is given below, courtesy of Felix Flicker, fellow of New College.



The key is that each clap reflects off a stair individually. When the echoes arrive back at the listener, they arrive quickly enough to be heard as a pitch.

The width and height of the steps are $w = 30\,\text{cm}$ and $h = 16\,\text{cm}$. Suppose one claps at a distance $L \gg w, h$. The path length differences for reflections off the bottom few steps are approximately $2w$, giving the frequency

$$f = \frac{v}{2w} = 570\,\text{Hz}$$

where we used $v = 343\,\text{m/s}$. The quack then continues, due to reflections off higher and higher stairs. Once the stairs are much further away than L, path length differences for subsequent reflections are approximately $2\sqrt{w^2 + h^2}$, giving frequency

$$f = \frac{v}{2\sqrt{w^2 + h^2}} = 500\,\text{Hz}.$$

Hence the quack consists of a pitch that starts high and then falls slightly lower as it fades away. For further discussion, see the article *How the Mound got its Quack*.

Remark

Phase shifts upon reflection for sound waves can be a bit tricky. Recall from **W1** that a hard boundary for a transverse string wave $y(x, t)$ sets y to zero. As a result, upon reflection, y flips sign, but $v_y = \partial y/\partial t$ stays the same.

When a sound wave hits a hard wall, the wall sets the displacement $\xi(x, t)$ to zero. Then upon reflection, the displacement flips sign, while the pressure variation $\delta P(x, t) \propto \partial\xi/\partial x$ stays the same. In standing waves, a hard wall is thus a node for ξ and an antinode for δP. Similarly, when sound waves in a tube reflect off an open end, the end sets δP to zero (since everything outside the tube has atmospheric pressure), so it flips sign. An open end is thus a node for δP and an antinode for ξ.

The rule is always the same: whatever quantity gets fixed to zero by the boundary gets flipped in sign upon reflection, and for a standing wave, that quantity has a node at the boundary. But it's confusing enough that some introductory textbooks get it wrong. Some even state, in their confusion, that "hard boundaries flip transverse waves but not longitudinal ones", which is definitely not true in general.

[4] **Problem 1.** Some conceptual questions about sound waves.

(a) What is larger for a sound wave, the relative density variations $\Delta\rho/\rho$ or the relative pressure variations $\Delta P/P$? Or does it depend on the situation?

(b) What is larger, the velocity of a sound wave v or the amplitude of the velocity variations Δu of the underlying particles? Or does it depend on the situation?

(c) A speaker periodically switches between emitting frequency f_1 and f_2, with a frequency f'. If you start moving, the f_1 and f_2 you hear will be Doppler shifted. Is f' also Doppler shifted?

(d) Get a coffee cup with a handle and tap on the rim with a spoon. You will hear two distinct pitches, e.g. if you tap directly above the handle, or 45° away from this point. Investigate what happens for different angles. Can you explain why this happens?

(e) According to introductory textbooks, the fundamental mode for a pipe of length L and radius $r \ll L$, closed at one end and open at the other, has wavelength $4L$. In reality, it's a little bit different because the radius is nonzero. Is the wavelength actually longer or shorter than $4L$?

(f) Find a way to produce beats in real life.

[3] **Problem 2.** In this problem, you'll work through Newton's slick derivation of the speed of sound. Instead of considering how parcels of gas move, we consider the force the gas exerts when squeezed.

(a) In **M4**, we showed that the speed v of longitudinal waves in a spring of length L, mass M, and spring constant k obeys $v^2 = kL^2/M$. For a cylinder of gas of length L and area A, show that the effective spring constant is

$$k = -A^2 \frac{dp}{dV}.$$

(b) Assuming the sound waves are adiabatic, use this to conclude that

$$v^2 = \frac{\gamma p}{\rho}.$$

If each gas molecule has mass m, rewrite the result in terms of γ, T, and m.

Next, we consider some limitations of this result.

(c) In an ideal gas, we assume the particles are noninteracting: they pass right through each other. But for sound waves to propagate, adjacent packets of ideal gas must exert pressure on each other. How is this possible? Use this observation to estimate the maximum possible angular frequency of sound in a gas in terms of the number density $n = N/V = p/k_B T$, the radius r of a gas molecule, and the speed of sound v.

(d) Our analysis also breaks down if the pressure variations are no longer adiabatic. The rate of heat conduction in a gas with thermal conductivity k_t across a surface of area A is

$$\frac{dQ}{dt} = -Ak_t \frac{dT}{dx}$$

For a sound wave with angular frequency ω, show that the adiabatic approximation holds when $\omega \ll pk_B/mk_t$. Does this hold for audible sound in air, where $k_t \approx 25\,\text{mW}/(\text{m} \cdot \text{K})$?

[2] **Problem 3.** A rubber rope with unstretched length L_0 is stretched to length $L > L_0$.

(a) Find the ratio of the speeds of transverse and longitudinal waves.

(b) Experimentally, it is found that the longitudinal waves are much more strongly damped. (You can check this at home, by making such a rope by tying together cut rubber bands.) Can you explain why, by considering the molecular structure of rubber?

[3] **Problem 4.** USAPhO 1998, problem B1. (The official solution has a qualitatively incorrect answer for the final part of the problem; see Stefan Ivanov's errata for the correct answer.)

[3] **Problem 5.** USAPhO 2016, problem A1.

2 Polarization

Now we'll introduce polarization for light waves, putting the results of **E7** to work.

Idea 2

The polarization of a light wave refers to the direction of its electric field; the light waves we saw in **E7** were linearly polarized. For example, a light wave traveling along $\hat{\mathbf{z}}$ with its polarization at an angle θ from the x-axis has electric field

$$E_x(z,t) = (E_0 \cos\theta)\cos(kz - \omega t), \quad E_y(z,t) = (E_0 \sin\theta)\cos(kz - \omega t).$$

A polarizer lets only light of a certain linear polarization through; if light with a linear polarization at an angle θ from this axis passes through it, then a fraction $\cos^2\theta$ of the energy is transmitted. Just as light can be incoherent, it can be unpolarized; unpolarized light hitting a polarizer loses half its energy.

[3] **Problem 6.** A simple polarizer contains many very thin, closely spaced wires. If the wires are vertical, they block vertical electric fields, allowing only horizontally polarized light to go through; this is a horizontal polarizer. One can similarly make diagonal and vertical polarizers.

(a) Suppose that perfectly monochromatic, but unpolarized light is incident on a double slit. (In this case, assume "unpolarized" means that at each instant in time, the polarization of the light passing through each slit is the same, but over longer timescales that polarization can vary.) What does the intensity pattern on the screen look like?

(b) Next, suppose a vertical polarizer is placed in front of one slit, and a horizontal polarizer is placed in front of the other slit. Now what does the intensity pattern look like?

(c) Finally, we further modify the setup of part (b) by placing many diagonal polarizers, at 45° to the vertical and horizontal, right in front of the *screen*. What does the intensity pattern on the screen look like now?

(d) On an unrelated note, suppose we wish to rotate the polarization of linearly polarized light by using $N \gg 1$ intermediate polarizers. What's the best way to do this, and what's the fraction of light that passes through the stack?

Idea 3

For a plane wave propagating along the z-axis with general polarization, it's useful to write

$$\mathbf{E}(z,t) = \text{Re}\left(\mathbf{E}_0\, e^{i(kz-\omega t)}\right)$$

where $\mathbf{E}_0$ is a complex two-component vector, describing both its amplitude and polarization. For example, if $\mathbf{E}_0 = E_0\,\hat{\mathbf{x}}$, the light wave is horizontally polarized, if $\mathbf{E}_0 = iE_0\,\hat{\mathbf{x}}$, it's horizontally polarized with a phase shifted by $\pi/2$, if $\mathbf{E}_0 = E_0\,\hat{\mathbf{y}}$ it's vertically polarized, and if $\mathbf{E}_0 = E_0\,(\hat{\mathbf{x}}+\hat{\mathbf{y}})/\sqrt{2}$ it's diagonally polarized.

When linear polarizations are combined with a relative phase, the result is circular (or more generally, elliptical) polarization. For example, when $\mathbf{E}_0 = E_0(\hat{\mathbf{x}}+i\hat{\mathbf{y}})/\sqrt{2}$, we have

$$E_x(z,t) = \frac{E_0}{\sqrt{2}}\cos(kz-\omega t), \quad E_y(z,t) = -\frac{E_0}{\sqrt{2}}\sin(kz-\omega t)$$

which is a circularly polarized light wave; the electric field at a fixed point rotates in a circle over time, and if one draws the electric field vectors in a line along $\hat{\mathbf{k}}$, they trace out a spiral. Birefringent materials, which have different indices of refraction in different directions, cause such phase shifts, and thus can convert linear polarizations into other polarizations.

Example 3

A plane wave with amplitude $\mathbf{E}_0 = E_0\,\hat{\mathbf{x}}$ enters a linear optical device, which does not absorb or reflect any energy. When the plane wave exits the device, it has circular polarization, $\mathbf{E}_0 = E_0\,(\hat{\mathbf{x}}+i\hat{\mathbf{y}})/\sqrt{2}$. What does the device do to light with vertical polarization?

Solution

Vertically polarized light has to exit with circular polarization of the other handedness, i.e. with $\mathbf{E}_0 = E_0 e^{i\theta}(\hat{\mathbf{x}}-i\hat{\mathbf{y}})/\sqrt{2}$ for some unknown phase θ, because this is the only possibility consistent with energy conservation.

To see this, note that the energy of a light wave is proportional to the time-averaged value of $|\mathbf{E}|^2$, which is in turn proportional to $|\mathbf{E}_0|^2$. Since horizontal and vertical polarizations are orthogonal, they don't interfere, so sending in both a horizontal and vertical light wave of amplitude E_0 at the same time just doubles the input energy. This must also double the output energy, and indeed, under the above ansatz we have

$$\hat{\mathbf{x}}+\hat{\mathbf{y}} \to \frac{\hat{\mathbf{x}}+i\hat{\mathbf{y}}}{\sqrt{2}} + \frac{\hat{\mathbf{x}}-i\hat{\mathbf{y}}}{\sqrt{2}} = \sqrt{2}\,\hat{\mathbf{x}}$$

which indeed has double the energy of one wave by itself.

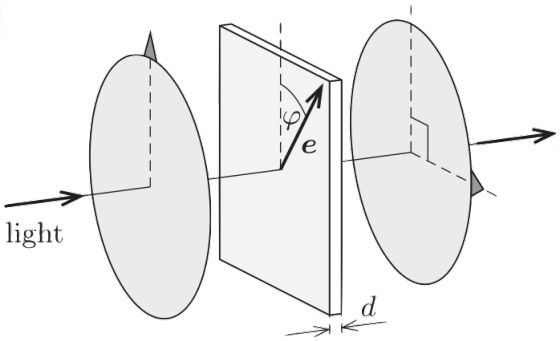
The more general principle here is that, since $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ were orthogonal to each other, they must be mapped to two other unit vectors which are still orthogonal, as complex vectors.

That is indeed true, because

$$(\hat{\mathbf{x}} + i\hat{\mathbf{y}})^\dagger(\hat{\mathbf{x}} - i\hat{\mathbf{y}}) = \hat{\mathbf{x}} \cdot \hat{\mathbf{x}} + i^2\hat{\mathbf{y}} \cdot \hat{\mathbf{y}} = 0.$$

Once we know what the device does to horizontally and vertically polarized light, we can find what it does to any polarization of light by superposition.

[2] **Problem 7** (MPPP 127)**.** A birefringent material is placed between two orthogonal polarizers. The material has thickness d, and has an index of refraction of n_1 for light linearly polarized along the axis $\mathbf{e}$, and n_2 for light polarized along an orthogonal axis.



If the system is illuminated with light of wavelength λ, give a value for d and orientation of $\mathbf{e}$ that maximizes the transmitted light.

[2] **Problem 8** (MPPP 128)**.** In the first 3D movies, spectators would wear glasses with one eye tinted blue and the other tinted red. This was quickly abandoned in favor of a system that used the polarization of light.

(a) If you wear an old pair of 3D movie glasses, close one eye, and look in the mirror, then you can only see the open eye. Explain how these glasses employ light polarization. What disadvantages might this system have?

(b) If you wear a new pair of 3D movie glasses and do the same, then you can only see the closed eye. Explain why.

[2] **Problem 9** (HRK)**.** A quarter-wave plate is a birefringent plate that causes a $\pi/2$ phase shift between light polarized along $\mathbf{e}$ and perpendicular to $\mathbf{e}$. Similarly, a half-wave plate causes a π phase shift. Suppose you are given an object, which may be a quarter-wave plate, a half-wave plate, a linear polarizer, or just a semi-opaque disk of glass. How can you identify the object? You can use an unpolarized light source, and any number of polarizers and quarter-wave and half-wave plates.

[2] **Problem 10** (HRK)**.** A polarizer and a quarter-wave plate are glued together so that, if the combination is placed with face A against a shiny coin, the face of the coin can be seen when illuminated by light of appropriate wavelength. When the combination is placed with face A away from the coin, the coin cannot be seen. Which component is on face A and what is the relative orientation of the components?

[2] **Problem 11.** Most optical elements are time-reversal symmetric, but the "Faraday rotator" is not. (It contains magnetic fields, and the direction of a magnetic field flips under time reversal.) When viewed from the top, linearly polarized light passing down through it will have its polarization rotated

clockwise by θ, but light passing up through it will *also* have its polarization rotated clockwise by θ. In other words, if linearly polarized light goes through a Faraday rotator, bounces off a mirror, and returns through it, it doesn't return with the same polarization, but rather is rotated by 2θ.

(a) Explain how to use a Faraday rotator, in combination with other optical elements mentioned above, to construct an optical isolator: a system which allows some light to pass through in one direction, but none to pass through in the other direction.

(b) In **T2**, we argued that a "one-way" light filter would violate the second law of thermodynamics. So why is the setup in part (a) allowed?

[4] **Problem 12.** IZhO 2021, problem 3. A problem on the propagation of light through a waveguide, unifying material from **E7** and **W1**.

3 Water Waves

Water waves are the most familiar examples of waves in everyday life, but you won't find them mentioned often in introductory textbooks, because they're far more complicated than any other kind of wave we'll consider. In all the problems below, we will completely neglect viscosity, surface tension, and compressibility of the water. Despite this, our results will still only be approximate.

[4] **Problem 13.** Consider water with depth D and density ρ. A shallow water wave (i.e. one with wavelength much greater than D) travels along the x-direction with height $h(x,t) \ll D$ relative to the water level. It turns out that the water has a horizontal velocity $v(x,t)$ which is approximately independent of height, and negligible vertical velocity. Even though the water is moving, the hydrostatic pressure formula still works because the water's vertical acceleration is negligible.

(a) Find a relation between the derivatives of $h(x,t)$ and $v(x,t)$ using conservation of mass. Using this result, show that the phase velocity v_w of the water waves obeys $v \ll v_w$.

(b) Find a relation between the derivatives of $h(x,t)$ and $v(x,t)$ using force and momentum.

(c) Combining these results, find the phase velocity v_w of shallow water waves.

Now let's consider what happens when a shallow water wave created at sea approaches the shore, and the depth D slowly decreases.

(d) Explain why waves always arrive at the shore moving perpendicular to the shoreline.

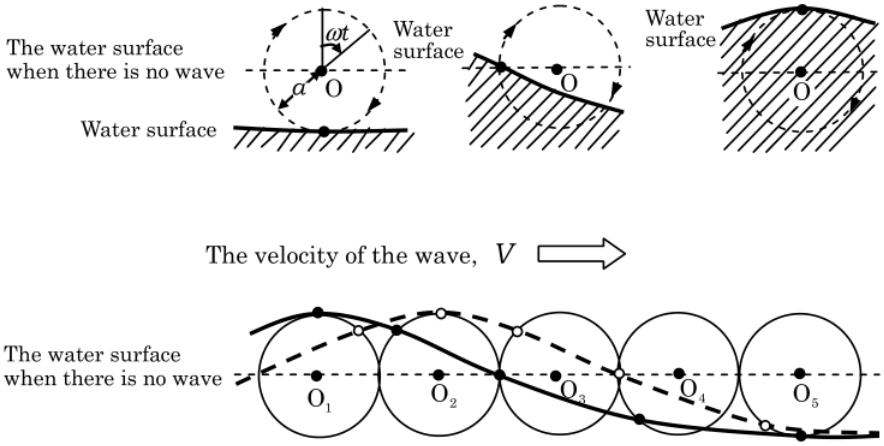
(e) If the depth is gradually halved, by what factor is the height of the wave multiplied? This phenomenon is known as shoaling.

Remark

Textbooks commonly say that liquids can't support transverse waves, because they don't support shear stresses. But the waves considered in problem 13 are clearly transverse. This is possible because the textbook statement only applies to the internal forces of water *alone*. At the surface of the water, gravity provides the transverse restoring force; that's why these waves are also commonly called "gravity waves".

[2] **Problem 14** (Japan)**.** In the above problem, we considered shallow water waves, $D \ll \lambda$, in which case the motion of the water is approximately horizontal. But in general, it turns out that the motion of the water will depend on height, and also that the individual water molecules move in ellipses. (This happens even when the wave has small amplitude, which we will always assume; the motion for large amplitude is even more complicated.) In the limit of deep water waves, $D \gg \lambda$, the water molecules move in circles. Using this fact, we can quickly derive the wave speed.

Assume the water molecules at the surface of the wave move in uniform circular motion with radius a and angular velocity ω, as shown.



(a) Consider the frame of reference moving to the right with velocity v. In this frame, the surface of the water is completely stationary, while molecules travel along the surface. Consider a small parcel of water which travels from a valley to a peak. By applying conservation of energy, derive a relationship between v, ω, and g.

(b) Find the phase and group velocity of the wave, in terms of g and the wavenumber k. In addition, find the condition on a and k for this derivation to make sense.

Showing that circular motion actually occurs takes more work, and involves solving partial differential equations; you can find a complete derivation here or in the first chapter here.

As you can see, water waves are quite complex. A diagram of the speeds of nine different limiting cases of water waves can be found in section 8.4 of The Art of Insight.

4 Reflection and Refraction

Now we'll introduce reflection and refraction with some real-world applications.

Idea 4

If a wave hits an interface, while traveling at an angle θ_1 to the normal to the interface, then it will generically both reflect and refract. The angle of the reflected ray is also θ_1, and the angle θ_2 of the refracted ray obeys $n_1 \sin\theta_1 = n_2 \sin\theta_2$. If there is no solution for θ_2 in the latter equation, then only reflection occurs.

These results follow directly from Huygens' principle, so they are very general, applying to light waves, sound waves, water waves, and so on, as long as the index of refraction n_i is always defined to be inversely proportional to the wave speed in each medium.

[1] **Problem 15.** Some conceptual questions about reflection and refraction.

(a) Does the index of refraction determine the phase velocity or the group velocity?

(b) Does a light beam of finite width get wider or narrower upon passing from air to water? Assume the light enters at an angle to the normal.

Example 4

Let the index of refraction at height h above the Earth's surface be $n(h)$. In terms of $n(0)$ and the Earth's radius R, what should dn/dh be at the surface so that light rays orbit in circles around the Earth, with constant height?

Solution

First, let's ignore the curvature of the Earth. Consider a light ray moving slightly upward, at a small angle θ to the horizontal, experiencing index of refraction n. Over a horizontal distance L, it goes up by a height $L\theta$. At this point, it will have a different angle θ' to the horizontal, and experience index of refraction $n + L\theta\, dn/dh$. Snell's law says

$$n \cos\theta = \left(n + L\theta \frac{dn}{dh}\right) \cos\theta'$$

and expanding to lowest order in the small angles θ and θ' gives

$$\frac{n}{2}\left(\theta'^2 - \theta^2\right) = L\theta \frac{dn}{dh}.$$

Approximating again to lowest order gives

$$\theta - \theta' \approx -\frac{L}{n}\frac{dn}{dh}.$$

Thus, the light ray turns through an angle of $(1/n)\, dn/dh$ per unit horizontal distance. For the light ray to stay at a constant height over the curved Earth, this must equal $1/R$, giving

$$\frac{dn}{dh} = -\frac{n(0)}{R}.$$

More generally, this calculation shows that light bends towards the direction with higher n. In the case of air, where $n - 1 \ll 1$, we can rewrite this as

$$\frac{d(n-1)}{dh} \approx -\frac{1}{R}$$

which can plausibly occur on Earth, due to the nice coincidence that $n - 1$ and H/R (where H is the typical scale height of the atmosphere) are both of order 10^{-3}.

Remark: Mirages

There are two classes of mirages.

- When $dn/dh < 0$, light rays bend down. If there is a distant object at the horizon, its image will appear *above* the horizon. This is called a "superior" mirage.
- When $dn/dh > 0$, light rays bend up. Then a distant object at the horizon will appear *below* the horizon, forming an "inferior" mirage. This also applies to the sky near the horizon, producing the illusion of water on the ground sometimes seen in deserts.

In air, the refractive index is close to 1, and $n - 1 \propto \rho \propto P/T$, where ρ is the air density and the second step used the ideal gas law. Usually we have $d\rho/dh < 0$, since $dP/dh < 0$ in hydrostatic equilibrium, but it depends on the value of dT/dh.

- In normal conditions, the Sun warms the ground and the hot air rises and adiabatically mixes the atmosphere (as discussed in **T1**), so that $dT/dh < 0$. This partially cancels the effect of the pressure variation, so that dn/dh is still negative but has small magnitude, so that mirage effects aren't apparent.
- In rare "thermal inversion" conditions, we have $dT/dh > 0$, so that dn/dh is negative with large magnitude, leading to strong superior mirage effects. If dn/dh is negative enough, it can match the value computed in example 4, allowing an observer to see arbitrarily far along the horizon despite the curvature of the Earth. This was the reason the famous Bedford Level experiment concluded the Earth was flat.
- In hot deserts, the air near the ground is very hot, so that $dT/dh < 0$ with a large magnitude. (A strongly negative dT/dh also occurs in cold days above water, since the water stays warmer than the air above it.) Here the temperature gradient overpowers the pressure gradient, so that $dn/dh > 0$ and inferior mirages can occur.

Proponents of the flat Earth hypothesis claim that the Earth only seems curved due to atmospheric refraction. But they have it backwards: in almost all conditions $dn/dh < 0$, which makes the Earth look *less* curved than it actually is.

[4] **Problem 16.** IPhO 1995, problem 2. Refraction in the presence of a linearly varying wave speed. (This is a classic setup with a neat solution, also featured in IPhO 1974, problem 2.)

[3] **Problem 17.** INPhO 2019, problem 1. Another exercise on refraction, with an uglier solution.

[3] **Problem 18.** USAPhO 2025, problem B2. A problem on shock wave wavefronts.

[3] **Problem 19.** IPhO 2003, problem 3B. An exercise on refraction and radiation pressure.

[4] **Problem 20.** IPhO 1993, problem 2. Another exercise on the same theme.

5 Ray Tracing

Idea 5

A pointlike object emits light rays in all directions. When those light rays subsequently converge at some other point, that point is the object's real image. If they don't actually converge, but all propagate outward with a common center, that point is the object's virtual image. In general, if we're given that an image exists, we can find its location by following the paths of selected rays from the object and looking for intersections.

[2] **Problem 21.** A pinhole camera is a simplified camera with no lens. It simply consists of a box with a small hole (the "aperture"). An image of the outside appears on the inside of the box (the "screen"), opposite the hole.

(a) Explain how the pinhole camera works by ray tracing.

(b) What are the disadvantages of having a larger or smaller aperture?

(c) Assuming the object being photographed is very bright, estimate the optimal aperture size for taking a clear picture with a pinhole camera, for a box of side length L.

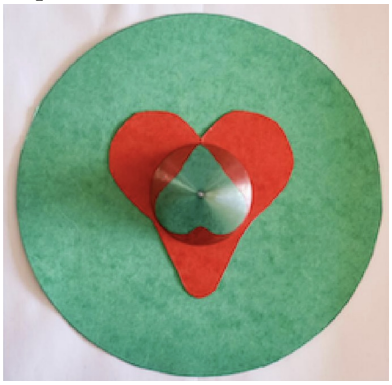
The next three problems will exercise your intuition with real-world examples.

[2] **Problem 22.** AuPhO 2020, section C.

[2] **Problem 23.** AuPhO 2013, problem 11. You'll also need the accompanying answer sheets.

[3] **Problem 24.** AuPhO 2019, problem 12. You'll also need the accompanying answer sheets.

[2] **Problem 25** (NBPhO 2024)**.** This photo shows the reflection of a red heart in a conical mirror.



The photo was taken from far above the mirror. In degrees, what is the mirror's apex angle?

[3] **Problem 26.** IZhO 2020, problem 1.3. A test of your intuition for 3D ray tracing.

Idea 6

Conic sections have some simple properties under reflection.

- Light rays emitted from one focus of an ellipse will all be reflected to its other focus.
- Light rays emitted from one focus of a hyperbola will all be reflected so that the resulting

rays all travel radially outward from the other focus.

- Parallel light rays entering a parabola along its symmetry axis (i.e. the axis perpendicular to the directrix) will all be reflected to its focus.

In the language of idea 5, if the foci of an ellipse/hyperbola are called F_1 and F_2, then an object at F_1 produces a real/virtual image at F_2. Note that a parabola is simply an ellipse in the limit where F_1 becomes very far away, so that rays coming in from F_1 become approximately parallel.

[2] **Problem 27** (Povey)**.** The mirascope is a toy consisting of two parabolic mirrors, pointing toward each other, so that the focus of each one is at the vertex of the other.



(a) When an object is placed at the bottom vertex, a real image appears at the top vertex. Why?

(b) How is the image oriented relative to the object?

The real image made by this setup is very convincing. There's a Michelin-starred restaurant that uses it in a course: when you reach for what looks like the food, your hand just passes through air.

Idea 7: Paraxial Approximation

If a light ray hits a thin lens of focal length f at a shallow angle, and at a distance $y \ll f$ above the lens's center, then it will exit the lens deflected by an angle $\pm y/f$, where the sign depends on whether the lens is converging or diverging. (For example, any light ray going straight through the lens's center isn't bent at all.)

This is the paraxial approximation, which only holds for light rays incident at shallow angles near the center of the lens. The quantity $P = 1/f$ is also called the optical power.

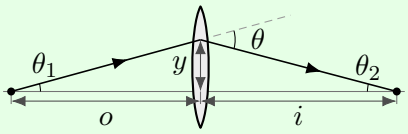
Conversely, if you don't know the focal length of a system, you can use this idea to find it. For example, the lensmaker's equation, giving the focal length of a lens of radii of curvature R_1 and R_2 and thickness d, is

$$\frac{1}{f} = (n-1)\left(\frac{1}{R_1} - \frac{1}{R_2} + \frac{(n-1)\,d}{nR_1R_2}\right)$$

and can be derived by computing the bending of the light ray at each interface.

Example 5

An object is placed a distance o behind a thin converging lens with focal length f.



An image is formed a distance i in front of the lens. How are o, i, and f related?

Solution

The horizontal light ray goes straight through, so let's consider another light ray which emerges at a small angle θ_1 to the horizontal. Then we read off

$$\theta_1 \approx \frac{y}{o}, \quad \theta_2 \approx \frac{y}{i}$$

but their sum is the deflection y/f, from which we conclude

$$\frac{1}{o} + \frac{1}{i} = \frac{1}{f}.$$

This is the familiar thin lens equation.

[2] **Problem 28.** Some basic questions about ray tracing.

(a) If two thin converging lenses of focal length f_1 and f_2 are placed right next to each other, what is the effective focal length f of their combination?

(b) Show that if you have height h, you can see your entire body in a vertical mirror of length $h/2$ whose top is at eye level, no matter how close or far you stand from it. Nonetheless, when people want to look at their whole body in the mirror, they typically stand back. Why?

(c) A candle is placed behind a converging lens. An image is formed on a screen on the other side of the lens. Now suppose that the *top* half of the lens is covered with a black cloth. Describe how the image changes.

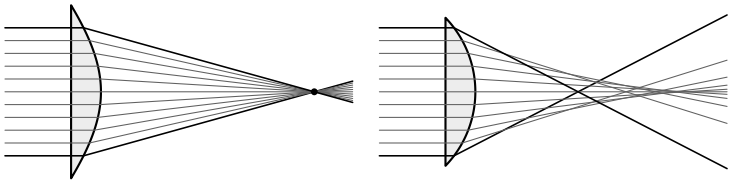
[3] **Problem 29.** USAPhO 2024, problem A3. A series of optics exercises relevant for real cameras.

[3] **Problem 30.** USAPhO 2026, problem A1. This question is straightforward, but assumes you know the definition $P = 1/f$ of optical power (expressed in "diopters", which just means m^{-1}).

Idea 8: Fermat's Principle

For fixed starting and ending points, light always takes the path of least time. This implies that if light from point P is all focused at point P', then all the relevant paths from P to P' take the same time. This principle is completely equivalent to the laws of reflection and refraction above, but may be more useful in certain situations.

[2] **Problem 31.** Parallel light rays coming in along the $+\hat{\mathbf{x}}$ direction enter a lens of index of refraction n, whose left edge is at $x = 0$ and whose right edge is described by the function $x(y)$. If all the light beams are to be focused at $x = f$, as shown at left below, what kind of curve does $x(y)$ have to be?



You should find that $x(y)$ is not an arc of a circle, which implies that a spherical lens will fail to focus all incoming horizontal light to a point. Instead, we will get spherical aberration, as shown at right above. However, most lenses are spherical because it's easier to make them that way.

The above questions only cover the most basic features of geometrical optics. For more practice on geometric optics in general, I strongly recommend Stefan Ivanov's collection of Russian problems. If you've taken a standard American high school physics class, you've probably already had some experience with lens systems, but if you want more, I recommend chapter 40 of Halliday and Resnick.